

Nanoseconds Reconfigurable Optical Switching Platform Based on REC-DFB Laser Arrays and AWGR

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Abstract *We demonstrate a nanosecond reconfigurable optical switching platform combining REC-DFB laser arrays, AWGR routing, FPGA-based scheduling, and visualized traffic steering to show destination-dependent wavelength selection and dynamic optical path reconfiguration in a four-node prototype.*
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Review Section

Recent data center and high-performance computing (HPC) workloads generate increasingly dynamic and latency-sensitive communication traffic, requiring interconnect architectures capable of high-capacity and fine-grained data exchange [1-4]. Conventional electrical packet-switched networks face growing challenges in scalability, latency, and power consumption. Optical switching has therefore attracted significant interest owing to its transparent transmission capability and potential to construct flatter network topologies with reduced electronic processing overhead.

Among various optical switching approaches, architectures based on Arrayed Waveguide Grating Routers (AWGRs) and reconstruction-equivalent chirp distributed feedback (REC-DFB) laser arrays have emerged as promising solutions for dynamic traffic environments [5,7,8]. Compared with conventional micro-electro-mechanical system (MEMS)-based optical switches, AWGR-based routing combined with nanosecond-scale wavelength switching enables faster optical path establishment and finer-grained traffic scheduling [5,6]. The AWGR is a passive wavelength-routing device without mechanically actuated switching elements in the optical core, and its cyclic routing property provides wavelength-dependent connectivity among input and output ports [9]. This makes it suitable for low-power all-to-all optical interconnection, while FPGA-assisted scheduling and synchronized wavelength control enable coordinated traffic forwarding and path reconfiguration.

This demonstration presents a nanosecond-scale reconfigurable optical switching platform integrating REC-DFB laser arrays, an AWGR, FPGA-controlled traffic controllers, and a centralized scheduler. It combines destination-dependent packet forwarding, flow-to-wavelength

mapping, synchronized wavelength switching, and real-time optical path reconfiguration. Visualization interfaces display traffic steering, wavelength assignment, AWGR-routed paths, optical spectra, and receiver-side signal quality, providing an experimental platform for low-latency and energy-efficient optical interconnects for future data center and HPC systems.

Novelty Section

The novelty of the demonstrated platform lies in the coordinated realization of nanosecond-scale optical switching across the optical hardware and system-control layers. Rather than presenting fast optical devices in isolation, the demonstration integrates wavelength switching, traffic forwarding, scheduling, and synchronization within a unified FPGA-controlled platform.

At the optical hardware layer, REC-DFB laser arrays and an AWGR establish optical paths through dynamic wavelength assignment [7,8]. The AWGR provides passive wavelength-dependent routing, where different wavelengths from a given input port are routed to different output ports, so destination changes can be realized by selecting the corresponding laser wavelength. Compared with MEMS-based approaches, this avoids mechanically reconfigured switching elements in the optical core and supports inherent all-to-all connectivity among nodes, making the architecture suitable for low-power, high-fan-out scenarios with frequent path changes.

At the system-control layer, FPGA-based traffic controllers perform packet parsing, buffering, forwarding, and flow-to-wavelength mapping, while a centralized FPGA-based scheduler coordinates wavelength selection and traffic transmission. FPGA-based time synchronization aligns the traffic controllers, scheduler, and wavelength-control circuits, allowing wavelength switching

and data forwarding to operate within a common reconfiguration process. This coordination enables system-level nanosecond switching, where each optical path change is matched with the selected traffic queue, destination node, and wavelength channel rather than evaluated only as an isolated device response.

ECOC Section

The ECOC community has long been interested in optical switching technologies for dynamic data center and HPC communication environments. While substantial progress has been achieved in fast optical devices, programmable optical networking, and low-latency interconnect architectures, the interaction between optical path reconfiguration, traffic scheduling, and wavelength routing is often difficult to observe directly in experimental systems.

This demonstration provides ECOC attendees with an opportunity to observe the operation of a nanosecond-scale reconfigurable optical switching platform beyond static performance results. Through visualization interfaces connected to the experimental prototype, attendees can follow how Ethernet traffic is classified by FPGA-based traffic controllers, how the selected destination determines wavelength assignment, and how the AWGR routes optical signals to the corresponding output ports. Optical-layer information, including laser spectra, optical power, wavelength selection, and receiver-side eye diagrams, is simultaneously displayed, allowing attendees to correlate wavelength selection with

network-level connectivity.

This simultaneous view of traffic control and optical-layer response is intended to make the reconfiguration process easier to interpret for ECOC attendees. Instead of observing only final connectivity or measured performance, attendees can examine how a change in destination is reflected in queue selection, wavelength assignment, AWGR routing, and received optical signals.

Demo Implementation Section

A. Demo Objectives

The demonstration presents a nanosecond-scale reconfigurable optical switching platform based on REC-DFB laser arrays and an AWGR. The platform establishes optical paths by coordinating traffic forwarding, FIFO-based buffering, wavelength selection, and AWGR-based passive routing. Its objective is to demonstrate a system-level implementation of fast optical path reconfiguration, where packet forwarding in the electrical domain and wavelength switching in the optical domain are jointly controlled to realize dynamic connectivity among network nodes.

B. Demo Configuration

The full picture of the Demo setup is as shown in Fig. 1. It shows a four-node optical switching prototype operating at 25 Gb/s per link, with Graphical User Interfaces (GUIs) illustrating the AWGR-based switching operation principle and optical signal status of the system.

The main functional blocks include the Traffic Controller, Network Scheduler, REC-DFB Laser

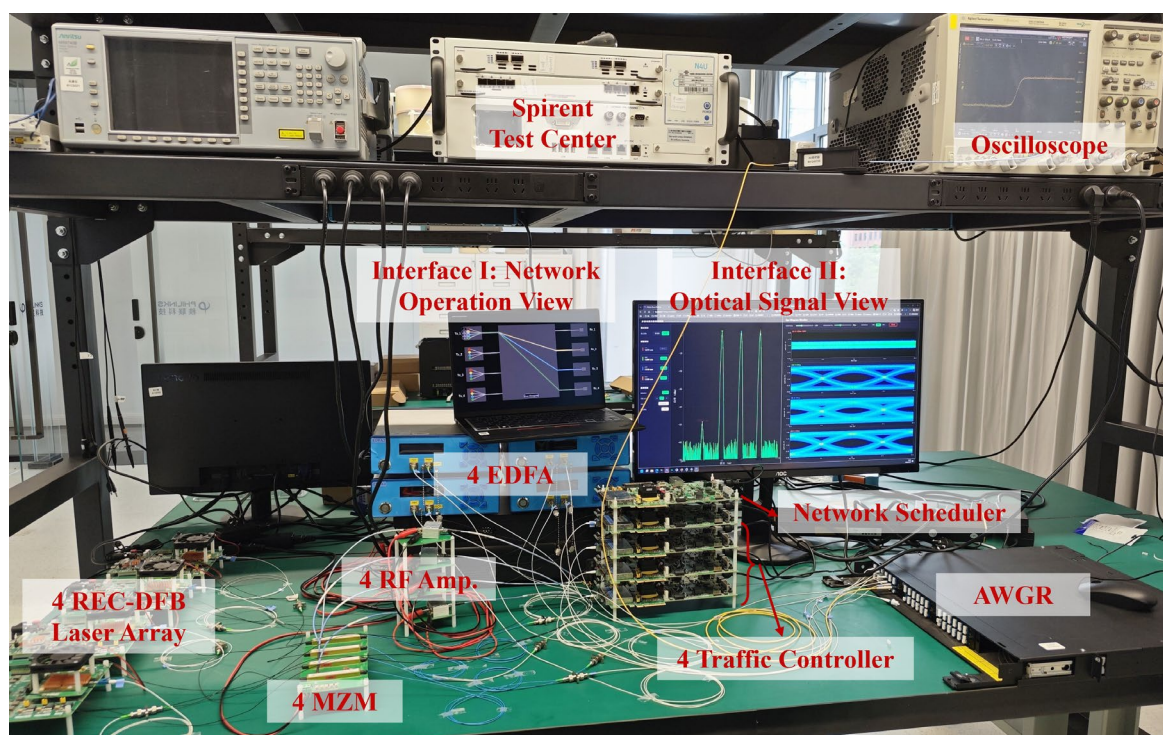


Fig. 1: Four-node optical switching prototype.

Array, Laser Driver, Mach-Zehnder Modulator (MZM), radio-frequency (RF) amplifier, erbium-doped fiber amplifier (EDFA), AWGR, and Spirent Test Center. The Traffic Controllers and Network Scheduler are implemented on AMD KCU116 FPGA boards. After destination-address identification, packets are assigned to different transmission queues according to their destination nodes. The Network Scheduler coordinates the transmission process and determines the corresponding wavelength assignment. The Laser Driver activates the selected channel of the REC-DFB Laser Array, and the modulated optical signal is routed by the AWGR to the target output port.

C. GUI Presentation & Attendee Interaction

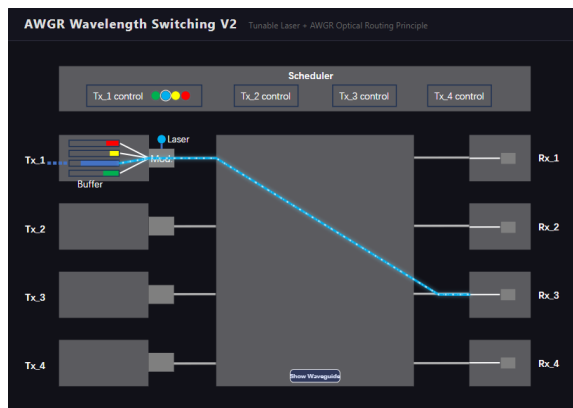


Fig. 2: Interactive Network Operation View.

The demonstration will be presented through interactive GUIs. First, a Network Operation View (Fig. 2) displays the four-node topology, source-destination connection, transmission-queue status, selected FIFO, wavelength-to-destination mapping, and AWGR routing path. This view illustrates how Ethernet traffic is classified according to destination information and associated with a specific optical path. It can be clicked to control the laser wavelength, triggering network switching status changes.



Fig. 3: Optical Signal View.

Another interface, referred to as the Optical Signal View and illustrated in Fig. 3, displays the output spectrum of the REC-DFB Laser Array, selected wavelength, and optical power level. Receiver-side eye diagrams, shown in Fig. 4, are

simultaneously displayed for multiple channels. The attendees can therefore clearly observe and understand the system dynamics in an end-to-end view. A pre-recorded *ns*-scale optical signal rising edge is also displayed, to further illustrate the fast switching capabilities of the AWGR-based system.

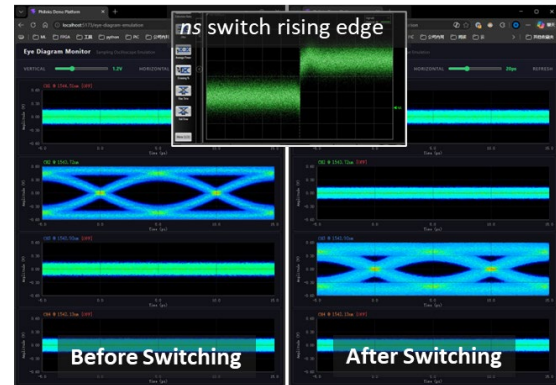


Fig. 4: Rising Edge and Receiver eye diagrams.

During the demonstration, presenters will first introduce the four-node switching architecture and the function of each hardware block. The audience can observe the GUI showing packets classified into destination-specific buffer queues. By manually clicking on the Network Operation View, the system can emulate buffer-threshold triggered switching operations and the corresponding automatic wavelength assignments. Once the selected wavelength is activated by the Laser Driver, the optical signal is launched into the AWGR and routed to the intended destination port. The Optical Signal View simultaneously displays the laser spectrum and receiver-side eye diagrams, allowing the switching process to be observed from both network-control and optical-signal perspectives.

Attendees can interact with the demonstration by modifying the communication destination through the GUI. After a destination is selected, the system updates the active transmission queue, wavelength assignment, and displayed AWGR routing path accordingly. The change in destination therefore leads to a corresponding change in the selected FIFO and laser wavelength, reconfiguring the optical path and directing the traffic to a different node.

Conclusions

This demonstration presents a nanosecond reconfigurable optical switching platform combining REC-DFB laser arrays, AWGR routing, FPGA-based control, and real-time visualization. It enables attendees to observe destination-dependent wavelength selection and optical path reconfiguration for low-latency data center and HPC interconnects.

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